

## Solution Requirements

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 15 July 2026                    |
| <b>Team ID</b>       | SWTID-2026-2567                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 4 Marks                         |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.N<br>o | Requirement<br>Category | Requirement Description                                                                            | Priority |
|----------|-------------------------|----------------------------------------------------------------------------------------------------|----------|
| 1        | User Input              | Allow users to enter applicant details through a web form.                                         | High     |
| 2        | Input Validation        | Validate all mandatory fields and data formats before submission.                                  | High     |
| 3        | Data Processing         | Convert user input into the format required by the ML model.                                       | High     |
| 4        | Prediction              | Predict whether the credit card application is Approved or Rejected using the Random Forest model. | High     |
| 5        | Result Display          | Display the prediction result to the user in a user-friendly format.                               | High     |
| 6        | Model Loading           | Load the trained machine learning model during application startup.                                | High     |
| 7        | Error Handling          | Display appropriate error messages for invalid inputs or system errors.                            | Medium   |
| 8        | User Interface          | Provide an easy-to-use and responsive web interface.                                               | Medium   |

### Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category                             | Requirement Description                                                                                                        | Target Metric / Acceptance Criteria                                                                                                                  |
|------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Performance &amp; Speed</b>           | The system should generate credit card approval predictions quickly after the user submits the application.                    | Prediction <u>result</u> displayed within <b>2–3 seconds</b> .                                                                                       |
| 2    | <b>Scalability</b>                       | The system should support multiple users accessing the application simultaneously without significant performance degradation. | Supports <b>100+ concurrent users</b> with stable performance.                                                                                       |
| 3    | <b>Security &amp; Data Privacy</b>       | The system should protect applicant information and validate all user inputs to prevent unauthorized access or invalid data.   | HTTPS deployment, server-side input validation, and secure storage of the ML model.                                                                  |
| 4    | <b>Reliability &amp; Availability</b>    | The application should provide consistent predictions and remain available during normal operation.                            | <b>99% uptime</b> with consistent prediction results for identical inputs.                                                                           |
| 5    | <b>Usability &amp; Accessibility</b>     | The application should have a simple, responsive, and easy-to-use interface for all users.                                     | User can complete an application in <b>under 5 minutes</b> ; compatible with desktop and mobile browsers.                                            |
| 6    | <b>Maintainability &amp; Portability</b> | The system should allow easy model updates and deployment on different platforms without major code changes.                   | <u>Model</u> can be replaced by updating the <u>.pk1</u> file; deployable on <b>Render</b> , <b>Railway</b> , or any Flask-supported cloud platform. |